VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY

UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY



COURSE: Introduction to Machine Learning (CSC14005)

MACHINE LEARNING PROJECT PROPOSAL

CLASSIFYING SIDEWALK PASSAGE OBSTRUCTION IN HO CHI MINH CITY FROM STREET-LEVEL IMAGERY

Instructor: PhD. Bùi Tiến Lên

Class: 24KHMT02

Academic year: Semester I, Academic Year 2026 - 2027

Group: 11

Students:

Full name	Student ID
Lê Nguyễn Quốc Học	24127044
Bùi Thị Bích Loan	24127072
Nguyễn Hữu Gia Minh	24127078
Nguyễn Lê Hoàng Khải	24127408
Trần Hoàng Phúc	24127505

Table of Contents

1 Problem Context and Motivation **1**
- 1.1 Practical problem . . . 1
- 1.2 Evidence and local relevance . . . 1
- 1.3 Gap to be addressed . . . 2

2 Machine-Learning Task and Project Scope **2**
- 2.1 Task formulation . . . 2
- 2.2 Data and application boundaries . . . 3
- 2.3 Planned data sources . . . 4
- 2.4 Experimental questions and evaluation . . . 4

3 Execution Methodology **5**
- 3.1 Step 1: Define and analyse the problem . . . 5
- 3.2 Step 2: Collect and prepare data . . . 6
- 3.3 Step 3: Preprocess and explore data . . . 6
- 3.4 Step 4: Select and train models . . . 6
- 3.5 Step 5: Evaluate and report . . . 7
- 3.6 Step 6: Build the application . . . 7
- 3.7 Step 7: Deploy and demonstrate . . . 7

4 Schedule, Team Organization, and Risk Control **8**
- 4.1 Planned schedule . . . 8
- 4.2 Team organization . . . 8
- 4.3 Risk register . . . 9
- 4.4 Expected deliverables and conclusion . . . 9

List of Tables

- 1 Published evidence motivating the project . . . 1
- 2 Label definitions and screening rules . . . 3
- 3 Project scope . . . 3
- 4 Planned roles and limitations of image sources . . . 4
- 5 Project milestones and required evidence . . . 8
- 6 Principal risks and planned responses . . . 9

1. Problem Context and Motivation

1.1. Practical problem

A sidewalk can exist physically yet fail as a walking route when parked vehicles, goods, tables and chairs, or temporary barriers break its continuous path. Pedestrians may then have to enter the roadway; a March 2026 report documented this situation for pupils near schools in HCMC [1]. Because encroachment can recur after enforcement, a single inspection cannot describe later conditions. This project therefore proposes an image-screening tool that records time-stamped observations and flags visible blocked paths for review.

The primary users are survey teams. During a route survey, the team collects street-level images; the system predicts whether the visible path is blocked; and a reviewer checks the image and metadata before selecting locations for field inspection. The initial evaluation focuses on classification quality; the effect on survey efficiency will be examined in future work.

1.2. Evidence and local relevance

Table 1 presents three directly relevant sources. Their measurements have different units and must be interpreted within their stated scopes.

Table 1: Published evidence motivating the project

Scope and date	Published evidence and interpretation
HCMC, reported 26 March 2026	An enforcement campaign processed 19,630 cases concerning roads, road margins, and sidewalks; the report also described re-encroachment after enforcement teams left [2]. The case count is neither a count of unique obstructed locations nor a sidewalk-obstruction rate.
HCMC school areas, reported 26 March 2026	A news report documented pupils entering the roadway where occupied sidewalks left no walking space [1]. This provides a concrete consequence, but it does not estimate how often the situation occurs citywide.
Central HCMC study, published online 11 July 2025	A study involving 381 child–parent pairs and audits of 88 walking routes found child-friendly walkability to be average or below [3]. It provides local context but does not directly measure the prevalence of sidewalk obstruction.

1.3. Gap to be addressed

The available evidence establishes a recurring local concern but does not identify which photographed segments have a usable path at a later survey time. The specific gap is a method for screening images for blocked passage, retaining reviewable evidence with capture metadata, and testing whether classification transfers to streets excluded from training. The proposed system supports prioritization; it does not replace field inspection or make legal or accessibility determinations. The closest prior resource is the Vietnamese Sidewalk Violation VQA dataset [4], whose question Q10 asks whether pedestrians still have a clear path. Because it lacks location and capture-time metadata, it cannot test transfer to unseen streets; this project adds geolocated, time-stamped imagery and a street-held-out evaluation. Comparable work exists elsewhere, such as Project Sidewalk [5]; this project targets a motorbike-dominated city.

The expected contributions are:

- **Dataset:** a locally verified, provenance-documented image set and annotation guide.
- **Model comparison:** at least three model families against a majority baseline.
- **Evaluation:** results on streets excluded from model training.
- **Application:** a web demonstration with a prediction API and reviewable observations.

2. Machine-Learning Task and Project Scope

2.1. Task formulation

The core task is supervised binary image classification. Each sample is one RGB image showing one side of a sidewalk; Mapillary panoramas will be converted into perspective views facing each sidewalk, oriented by the direction of travel, and other images showing both sides will be cropped by a documented rule before annotation. The label applies only to the visible segment at capture time. Location and time are retained as metadata for splitting and review, not used as prediction features.

Let x_i be an image and $y_i \in \{0, 1\}$ its verified label. The model returns a decision score, and the validation set determines the threshold:

$$s_i = f_\theta(x_i), \qquad \hat{y}_i = \mathbf{1}[s_i \geq \tau]. \tag{1}$$

This notation does not assume that a decision score is a calibrated probability. Annotators apply one decision test: would a pedestrian have to step into the roadway to continue along the

visible segment? Table 2 defines the resulting labels and the separate screening state.

Table 2: Label definitions and screening rules

Label or state	Observable rule and examples
0: Visible path remains	A continuous strip of unobstructed sidewalk surface is visible through the photographed segment. A motorcycle parked near the edge while a continuous path remains is label 0. This label does not certify a compliant width or universal accessibility.
1: Blocked	An obstacle cuts across the walking path and no continuous clear strip remains in the visible segment. A row of vehicles or goods occupying the full path is label 1; merely seeing a vehicle or stall is insufficient.
Insufficient information	Used only when the image lacks the evidence to answer the decision test: the sidewalk area is not in view, the image is blurred or badly framed, or the decisive part is occluded. A sidewalk fully covered by obstacles is labelled blocked, and difficult but visible cases still receive 0 or 1. Unresolved items are excluded with a recorded reason; this is not a third model class.

2.2. Data and application boundaries

Table 3: Project scope

In scope	Out of scope
Approximately ten HCMC streets; dry daytime images; passage blocked by parked vehicles, goods, tables and chairs, or barriers; binary image classification; and stored observations with genuine source, place, and time metadata when available.	Measuring clearance in metres; producing a 0–100 score; counting or localizing individual obstacles; comprehensive assessment of surface quality or accessibility; real-time monitoring; citywide prevalence estimates; and legal conclusions.

The final street list will be fixed after the pilot and will not be presented as representative of HCMC. In the application, reviewers can mark an upload as insufficient information before or after viewing a prediction and exclude it from the observation list. The binary classifier is not presumed to recognize every out-of-scope input, so input guidance, validation, and human review remain necessary. A map shows recorded observations rather than live conditions, and missing coordinates never receive fabricated locations.

2.3. Planned data sources

Table 4: Planned roles and limitations of image sources

Source	Planned role, metadata, and labels	Key limitation
Mapillary [6]	Planned primary source for street-grouped training, validation, and testing, and for mapping observations if the pilot confirms usable imagery, coordinates, capture dates, attribution, and access. All selected images will be annotated under this project's guidelines.	Availability and sidewalk visibility may vary by street; usage conditions must be checked before collection or release.
Team-captured images	At least 100 planned images for new-source evaluation and demonstration, with street, capture session, time, and coordinates recorded when safe and available. Labels will be independently reviewed.	Images from a street used in training test only new images on a known street; they are not evidence of unseen-street transfer.
VQA Q10 [4]	Optional out-of-domain check on a relabelled subset; not used for training. Before use, the team will verify files, provenance, licence, and any usable metadata.	No location or capture-time metadata, so these images cannot enter the map or the main street-based experiment.

The planned target is 1,500–3,000 non-duplicate images, including at least 100 team-captured images, subject to revision after a 100–200 image pilot. These are targets, not collected-data counts. Original images or panoramas will be distinguished from derived crops, and all crops from one parent image will remain in the same split.

2.4. Experimental questions and evaluation

The primary question is: *Which planned model best classifies sidewalk passage obstruction on streets excluded from training?* A secondary analysis will examine performance on a new image source. A team-captured image from a training street will be reported as a known-street

result, while an image from a held-out street may contribute to unseen-street evaluation if its provenance is verified. VQA Q10 images, if used, will be reported only as a separate out-of-domain check.

Training, validation, and test sets will be grouped by street; all crops from one panorama, each team capture session, and each near-duplicate group will stay within one split. The street list will be chosen so that the held-out test contains several streets. The report will give image counts by class and street, plus per-street results when sample sizes permit.

Each candidate model will be trained and tuned using only the training and validation sets. After validation-based tuning, each candidate's configuration and decision threshold will be frozen. All candidate models and the majority-class baseline will then be evaluated once on the same held-out test set. The deployment model will be selected using validation results; test results will support final comparative reporting rather than further model selection or tuning.

Macro-F1 is the primary metric:

$$F_{1,\text{macro}} = \frac{1}{2} \sum_{c \in \{0,1\}} \frac{2\,\text{Precision}_c\,\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \tag{2}$$

Per-class precision and recall, confusion matrix, sample counts, and a majority-class baseline will also be reported, together with the share of collected images that are eligible and the reasons for exclusion. Street-level bootstrap confidence intervals and PR-AUC or Average Precision are conditional additions if enough independent streets and positive cases are available. No performance target is guaranteed before the pilot and final test.

3. Execution Methodology

The seven stages below describe planned work rather than completed experiments. Each stage produces a versioned artefact and must pass its gate before dependent work proceeds.

3.1. Step 1: Define and analyse the problem

The team will finalize the survey question, annotation guide, and candidate streets. Two annotators will label a 100–200 image pilot and revise unclear boundary rules through documented adjudication.

Output: problem specification, survey scope, and annotation guide version 1.

Gate: recurring pilot disagreements have explicit rules before full collection begins.

3.2. Step 2: Collect and prepare data

Images will be sampled by source and street toward the planned scale in Section 2. For each item, the manifest will record source or URL, parent-image ID, street, capture session where applicable, available capture time and location, usage conditions, annotator, and review state; preliminary labels will follow the common guide.

Cropping and duplicate detection will be automated, but every label will be assigned manually: validation and test images blind by two annotators without model suggestions, and training images by one annotator with a cross-checked sample. The pilot will observe the share of eligible images and the labelling time per image to size the screening effort.

Output: traceable image repository, manifest, collection log, and preliminary labels.

Gate: actual counts by source, class, and street are inventoried and ineligible or unknown-location items are separated.

3.3. Step 3: Preprocess and explore data

The team will remove corrupt or unusable files, detect exact and near duplicates, keep related crops together, and review privacy-sensitive content. Independent annotation of the validation and test sets and a training subset will be summarized with raw agreement and Cohen's κ before adjudication; EDA will then describe class, street, source, capture-time, and quality distributions. Images removed by any automatic pre-filter will be spot-checked so that blocked cases are not disproportionately lost. Splits will be grouped by street, keeping parent images, team capture sessions, and near-duplicate clusters together.

Output: cleaned data, final guide, EDA notebook, and fixed split lists.

Gate: both test classes are represented; no street, team capture session, parent image, or near-duplicate group spans multiple splits; and the test set is frozen.

3.4. Step 4: Select and train models

Using identical splits, the team will compare at least three model families against a majority-class baseline, each chosen to answer a different question: a classical classifier on hand-crafted features (e.g., HOG with an SVM) tests how far simple visual cues go; a fine-tuned CNN (e.g., ResNet-18) tests end-to-end learning on local data; and frozen pretrained embeddings with a linear classifier (e.g., CLIP) test transfer from large-scale pre-training. Specific architectures and configurations will be fixed after the pilot according to compute resources. Each model's hyperparameters, class weighting, and decision threshold will be selected on validation data and frozen before test evaluation; learning curves will be inspected for overfitting, and seeds, preprocessing, versions, configurations, and logs will be retained.

Output: trained candidates, frozen configurations and thresholds, and a validation comparison.

Gate: every candidate is reproducible and frozen before the common test set is opened.

3.5. Step 5: Evaluate and report

All frozen candidate models and the majority-class baseline will be evaluated once on the same held-out test set using the metrics in Section 2; test results will not trigger further selection or tuning. Recall for the blocked class will receive particular attention, because a missed blocked image is a location that escapes review. Error analysis will examine cases such as vehicles beside a remaining path, narrow gaps, occlusion, and difficult viewpoints. Source- and street-specific results will be reported only with adequate samples, and conditional confidence intervals or ranking metrics will be clearly marked.

Output: shared-test comparison, error analysis, and validation-selected deployment model.

Gate: no test-driven refinement is reported as final; any such result requires a new test set or an exploratory label.

3.6. Step 6: Build the application

A FastAPI service and web interface will support image upload, score and label display, manual-review status, and an observation list or map using genuine metadata. Users can also classify a permitted Mapillary image retrieved with its capture metadata through the Mapillary API. API credentials will remain on the backend, and source failures will produce a clear status message; the flow will not crawl the city or imply real-time monitoring.

Output: web interface, prediction API, Mapillary selection flow, and user guide.

Gate: valid inputs match offline predictions, invalid inputs fail clearly, and missing locations create no map points.

3.7. Step 7: Deploy and demonstrate

The team will deploy the web/API publicly with pinned dependencies, environment settings, and a visible model version. The demonstration will include new images, errors, review-required cases, latency, and an offline fallback.

Output: public deployment and complete submission package.

Completion criterion: an independent user can open the URL, run the main flow, and verify the deployed model version.

4. Schedule, Team Organization, and Risk Control

4.1. Planned schedule

Table 5: Project milestones and required evidence

Milestone	Primary work	Required evidence
Week 3	Finalize the problem, scope, and annotation guide; conduct the Steps 1–2 pilot; initialize the web/API scaffold with sample records.	Proposal; pilot and feasibility logs; interface skeleton; unresolved-item list.
By Week 6	Complete collection, annotation, preprocessing, EDA, and data splitting.	Image manifest; annotation guide; EDA; frozen split lists.
Weeks 7–10	Train and tune at least three model families; select the deployment model on validation; freeze all candidates; evaluate all models and the majority baseline once on the same test; integrate the selected model into the API.	Logs and frozen configurations; shared-test metric table; confusion matrices; selected model; offline/API parity check.
Reporting week	Complete cloud deployment, demonstration, report, and slides.	PDF report; source-code archive; saved model; slides; public URL.

4.2. Team organization

Each member will lead one workstream:

1. Problem definition and source verification
2. Annotation, EDA, and data splitting
3. Baseline and model development
4. Web and API development
5. Integration, deployment, and reporting

All members will cross-check outputs, and at least two will annotate independently. The model lead cannot change test labels after viewing predictions. Final assignments will be agreed by the team.

4.3. Risk register

Table 6: Principal risks and planned responses

Risk	Planned response
Too few valid images or severe class imbalance	Test feasibility in the pilot; add streets, self-captured images, and minority-class cases where needed. Report the actual achieved scale and never substitute near duplicates for independent observations.
Ambiguous labels or data leakage	Revise the guide, use independent adjudication, and group by street, keeping team capture sessions, parent images, and near-duplicate clusters together. Use only verified-street images in the main experiment and keep the test set sealed.
External API access or usage restrictions	Verify access, licence conditions, and attribution during the pilot. Prefer team-captured images when an external source is unsuitable; review personal information and keep a compliant offline demonstration set.
Model overfitting or weak street transfer	Monitor validation learning curves, keep street-disjoint splits, limit tuning to validation data, and report subgroup and error analyses.
Deployment or external-service outage	Pin dependencies, record the model version, validate the offline/API parity, and prepare a fallback demonstration that does not fabricate live data.

4.4. Expected deliverables and conclusion

The final submission is expected to contain a verified image manifest, annotation guide, EDA and experiment notebooks, reproducible configurations, trained-model artefacts, a report and presentation, source code for the web interface and API, and a working deployment URL. Claims will remain proportional to the evidence: the map will describe recorded observations, evaluation will describe the sampled streets and conditions, and the classifier will be presented as a survey-support tool rather than a legal or accessibility certification system. Observations will be reported at street level, without identifying individual vendors or people.

References

[1] Thanh Nien Newspaper. What does the Ho Chi Minh City department of construction say about pupils having no sidewalk on which to walk? Published 26 March 2026, article, March 2026. Accessed 18 October 2026.

[2] Kim Sáng. Ho Chi Minh City imposes more than VND 17 billion in fines for sidewalk violations. Công Lý, 26 March 2026, article, March 2026. Accessed 18 October 2026.

[3] Thi Hanh An Le, Tu Anh Trinh, An Neven, Davy Janssens, and Quang Ngoc Truong. Child-friendly walkability for well-being in Ho Chi Minh City, Vietnam: An integrated approach from objective and subjective perspectives. *Cities & Health*, 2025. Published online 11 July 2025; doi:10.1080/23748834.2025.2520113; accessed 18 October 2026.

[4] Manh Dung Dao and Thanh Huy Cao. Vietnamese Sidewalk Violation VQA: A question-answering dataset for sidewalk encroachment detection. SE365 course project, University of Information Technology, VNU-HCM, 2026. Dataset card; accessed 18 October 2026.

[5] Galen Weld, Esther Jang, Anthony Li, Aileen Zeng, Kurtis Heimerl, and Jon E. Froehlich. Deep learning for automatically detecting sidewalk accessibility problems using streetscape imagery. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '19)*, 2019. doi:10.1145/3308561.3353798; accessed 18 October 2026.

[6] Mapillary. An introduction to Mapillary, 2024. Mapillary documentation; accessed 18 October 2026.